



Unseen seeds, green minds in the digital era: How SMEs foster innovation through green intellectual capital, AI, and resource commitment

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ABSTRACT

The rapid acceleration of digital transformation has profoundly reshaped societal operations, with Artificial Intelligence (AI) and the Metaverse emerging as key drivers of change. The strategic integration of tacit knowledge and advanced technologies is crucial for enhancing innovation performance and maintaining a competitive advantage in the global economy. This study explores how tacit knowledge sharing influences innovation performance (IP), with a focus on green intellectual capital (GIC), AI, and resource commitment (RC). To examine these relationships, a survey was conducted in three phases among small and medium-sized manufacturing firms in Canada and the UAE, collecting data from 317 UAE and 383 Canadian SMEs. The findings show that GIC plays a key role in linking tacit knowledge sharing to IP, while AI influences how tacit knowledge interacts with GIC, which consists of green human, relational, and structural capital. Additionally, RC strengthens the connection between GIC and IP, reinforcing the importance of resource investment in sustainability and innovation. The originality of this study comes from exploring how AI influences the connection between tacit knowledge and GIC, as well as how RC impacts the link between GIC and IP. The study also recognizes its limitations and outlines areas for future research, encouraging further research into the role of knowledge and technology in driving innovation.

Notation list

AI	Artificial intelligence
GIC	Green intellectual capital
TK	Tacit knowledge sharing
IP	Innovation performance
RC	Resource commitment
KBV	Knowledge based view
TA	Tacit Knowledge awareness
GSC	Green Structural Capital
GHC	Green Human capital
GRC	Green relational capital

1. Introduction

The digital transformation has fundamentally reshaped daily life, influencing how individuals live, work, and how policies are formulated and implemented (Al-Emran, 2023; Melović et al., 2020). This transformation has given rise to two prominent technologies: Artificial Intelligence (AI) and the Metaverse (Jafar et al., 2023; Kautish et al., 2025). To effectively adapt to these changes, it is crucial to embrace these technologies rather than resist them, as they form the foundation of new knowledge-based strategies for the evolving “new normal” (Alinasab et al., 2022; Grybauskas et al., 2022; La Sala et al., 2024). Firms increasingly recognize tacit knowledge as a vital resource for

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driving innovation, which serves as a cornerstone of developed economies. The advancement of these economies is often closely linked to their capacity to generate knowledge and cultivate creativity. Consequently, understanding the impact of tacit knowledge on innovation performance is crucial, particularly in the era of technological advancements such as AI and the Metaverse. Multiple studies underscore the importance of enhancing innovation performance as a key factor in achieving competitive advantage (Liu et al., 2023; Patel et al., 2024). Innovation performance, defined as the capacity of firms to develop and introduce new or improved green products, services, and processes, is a vital requirement for success in today's manufacturing environments (Ganguly et al., 2019; Li et al., 2023). This study will thus investigate deeper into these dynamics, investigating how firms can enhance their innovation performance through the integration of tacit knowledge, resource commitment, and the application of artificial intelligence.

The convergence of AI and the Metaverse offers transformative opportunities for fostering innovation by facilitating new forms of collaboration and knowledge exchange. These technologies provide platforms for organizations to unlock and disseminate tacit knowledge, which is crucial for driving creativity and enhancing innovation performance in increasingly competitive markets (Le et al., 2024). While numerous studies have explored the influence of various factors on innovation, one often-overlooked element is tacit knowledge. This information, inherent in individuals' cognitive processes and real-world experiences, has significant potential to enhance creativity (Kautish et al., 2024; Kodama, 2019). Enhancing individuals' understanding and awareness of tacit knowledge is a crucial first step in efficiently using tacit knowledge for innovation. Innovation fundamentally relies on the creation and sharing of knowledge, with the exchange of tacit knowledge among members being particularly crucial for enhancing organizational innovation (Acharya et al., 2022; Ardito et al., 2022). Numerous organizations have cultivated innovative ecosystems underpinned by such knowledge exchange. For instance, Google actively organizes developer conferences and forms online forums and communities (<https://developer.android.com>) to foster an environment conducive to the exchange of tacit knowledge. Nevertheless, much of this valuable knowledge remains elusive, embedded in the minds of individuals (Enwereuzor, 2021). Organizational scholars emphasize that knowledge sharing is a critical component of knowledge management, offering firms a sustainable competitive advantage and promoting effectiveness (Obrenovic et al., 2020). Tacit knowledge is particularly vital, as a significant portion of valuable knowledge within organizations is often tacit (Holste & Fields, 2010). This form of knowledge, deeply rooted in individual experience, skills, and expertise, is challenging to express or replicate (Mishra & Kautish, 2025; Smith, 2001) and is primarily acquired through social interaction, communication, and observation (Siddiqui et al., 2023).

AI and the Metaverse are transforming how institutions leverage GIC by building immersive knowledge sharing and collaboration environments (Raman et al., 2025). These tools make tacit knowledge more accessible and useable, which stimulates the creation of green relational, structural, and human capital and, in turn, triggers better innovative firm performance (Pereira et al., 2024). GIC plays a bridging role in linking tacit knowledge and innovative outcomes. GIC relates to green relational capital, refer to connections and networks; green structural capital, which refers to processes, systems, and facilities and green human capital, referring to skills and knowledge of employees (Rastogi, 2002; Shahbaz et al., 2023). According to Barney (1991), businesses could foster a workforce that is inclined to innovate by investing in green human capital development programs that enable the sharing and understanding of tacit knowledge. The development of green relationship capital through networks assists in the transfer of tacit knowledge among teams. Green structural capital eases the access of green innovation projects teams to tacit information by means of recording, storing and retrieval of tacit knowledge through efficient knowledge management systems and process (Bamel et al., 2022; Zhang et al., 2025).

In the industrial sector, AI is more frequently employed to amplify human capabilities and reveal untapped potential for information (Chopra et al., 2024; Carayannis et al., 2017). As Jiménez-Jiménez and Sanz-Valle (2011) argue, the AI can be used to automate work processes so that one can concentrate on other creative and strategic pursuits. In addition, AI can delve into copious amount of information for potentially revealing insights and buried patterns in tacit knowledge, which may trigger to develop new ideas and solutions (Kodama, 2019). Resources should be allocated for AI to be successful in an organization. The resources have funding for efforts around change management, infrastructure and training. RC as a reflection of an organization's activities into organizational strategies is considered part of strategy success planning (Cui & Lui, 2005). Previous studies have shown that RC has a strong effect on knowledge management, as high-performing firms tend to benefit from the alignment among information technology resources and resource commitment, which are critical resources of the organizational context (Appiah & Essuman, 2024). Barham et al. (2020) suggest that firms are encouraged to innovate by seeking new knowledge or finding novel ways to utilize their existing knowledge. Organizations must harness both tacit and explicit knowledge to generate new insights, which can ultimately lead to innovative products or services (Bamel et al., 2021). Rooted in the KBV, this study emphasizes the critical significance of tacit knowledge in fostering innovation and explores three interrelated research questions: (1) To what extent does the tacitness of knowledge drive innovation performance through its influence on green intellectual capital (GIC)? (2) How does AI act as a moderating mechanism in the relationship between tacit knowledge sharing and GIC? (3) In what ways does RC enhance the relationship between GIC and innovation performance by strengthening the integration of green intellectual assets into organizational processes?

This study addresses critical gaps in the KBV literature by integrating AI and RC as moderating factors in the relationship between tacit knowledge sharing, GIC, and innovation performance. While the KBV traditionally emphasizes the strategic importance of knowledge as a competitive asset (Grant, 1996), this study advances the paradigm by demonstrating how AI facilitates the capture, codification, and dissemination of tacit knowledge, thereby enabling the development of GIC. By focusing on manufacturing SMEs in Canada and the UAE, two culturally diverse and economically distinct contexts, the research extends the KBV framework into the realm of sustainability and cross-cultural knowledge management. The research provides new insights on the moderating effect of AI in translating tacit knowledge into actionable information leading to improvement in GIC. Expanding Nonaka and Takeuchi (1995) seminal work highlighting tacit knowledge as a cornerstone to organization knowledge creation, this study showcases how AI facilitates knowledge sharing in widely different organizational environments. For instance, Canadian SMEs leverage AI-powered collaboration tools to formalize and codify tacit knowledge, aligning with their low-context, individualistic-communication culture. On the contrary, UAE SMEs exploit AI-based informal knowledge sharing systems, which is due their high-context, collectivist society. The results contribute to the KBV by showing why various culture affect the mechanisms of knowledge management and innovation performance. The study thus, contributes by filling this gap by examining how AI and knowledge-based assets shape imitation and innovation, highlighting the relevance of KBV in addressing longstanding strategic challenges faced by firms.

Furthermore, this paper uncovers RC as a vital boundary condition, to amplify the relationship between GIC and innovation performance, which supports Barney's (1991) argument that resource allocation is the foundation of strategic advantage. By exploring how RC leads manufacturing SMEs to leverage AI use and sustainable practices, this research provides empirical evidence of the value of aligning organizational resources with knowledge management objectives. This study would support that Canadian and UAE SMEs could choose different levels of RC when using GIC to improve innovation procedure. Moreover, by embedding sustainability agendas and technology-induced

change within the KBV paradigm, this paper makes a new and integrated contribution to understanding how firms can exploit knowledge for sustainable innovation and competitiveness over time.

2. Theoretical background and hypotheses development

The knowledge-based view (KBV) paradigm builds upon foundational management theories such as stakeholder theory and the resource-based view (RBV) of the firm. KBV posits that knowledge generated within an organization is a crucial resource for achieving sustainable competitive advantage in dynamic environments (Cooper et al., 2023) and is often regarded as an extension of the RBV paradigm. This is because knowledge-based resources are challenging for organizations to understand and socially integrate, difficult for competitors to imitate, and are continuously evolving and co-creating (Chowdhury et al., 2022). KBV views organizations as accelerators and producers of knowledge. According to KBV, businesses are knowledge-integration entities, and their primary competency lies in structuring processes that enable employees' specialized knowledge to be utilized effectively (Al Dulaimi et al., 2022). The KBV emphasizes the importance of structural and systemic traits that allow businesses to develop the capability to connect complementary and specialized data. This shift from RBV to KBV highlights the significance of knowledge within individuals and the organization as a whole, rather than relying solely on physical resources for competitive advantage.

The most critical strategic resource, according to KBV, is implicit knowledge, also known as intellectual capital. It explores basic issues related to the firm theory, comprising organizational structure, the management roles, innovation theory, and coordination of knowledge within the firm (Grant, 1996). For a firm, tacit knowledge and GIC are the resources considered most important for strategy. Many businesses now rely on AI and the Metaverse to use knowledge-based resources in a better way (Shen et al., 2022), and companies such as Siemens use AI-based simulations in the Metaverse to make industrial manufacturing more efficient. They make it easier for firms to combine tacit knowledge and intellectual capital which improves their chances of achieving unique advantages in the market. By connecting AI and Metaverse with GIC, simplify the process of sharing tacit knowledge and make better use of resources (Umar et al., 2023), which aligns with KBV theory. In this study, KBV is applied to analyze the ways that TS impacts a firm's innovation performance using green intellectual capital. Furthermore, investigates the way AI connects with TS and GIC, as well as the interaction RC has with GIC and firm IP (see Fig. 1).

2.1. Tacit knowledge awareness (TA) and tacit knowledge sharing (TS)

Tacit knowledge, while difficult and costly to replicate, is frequently associated with innovation (Kogut & Zander, 1992). However, its complexity poses significant challenges to knowledge sharing within organizations (Fleck, 1996). Innovation is undeniably linked to improved corporate performance, particularly in knowledge-intensive industries (Jiménez-Jiménez & Sanz-Valle, 2011). Research has shown that effective tacit knowledge management can enhance employee performance and organizational skills, thereby positively impacting organizational performance (Lee & Choi, 2003). Tacit knowledge, generated and utilized by employees in their routine works, is critical for productivity improvement in manufacturing settings (Nakano et al., 2013). It involves intuitive and unconscious hunches and insights that include both how-to and know-why, so it is more intuitable than describable.

The "onion" model provides a more integrative way to understand tacit knowledge and argues for the existence of multiple levels of tacit knowledge associated with different tacitness levels. It describes three layers: a demonstrated tacit layer, a tacit reflecting layer, and a hidden practical layer (Asher & Popper, 2019). This model goes beyond the recognition of tacit knowledge in the classical typology and shows it as a multi-dimensional phenomenon related to varying levels of awareness. Showing the links between these layers in the onion model makes it easier to study tacit knowledge in detail. Finding out new things in one layer can show tacit information in other layers more apparent, highlighting to comprehensively explore the implicit knowledge within a particular profession, such as sales (Hedlund et al., 2002). The onion model differs from other models by recognizing tacit knowledge in general and its different levels, instead of focusing on terminological dichotomies (Collins & Evans, 2019). The techniques and concepts are useful in both practice and research, suitable for all groups and address the link between numerous layers of tacit knowledge. Tacit knowledge awareness refers to the recognition of new insights or ideas, often emerging as implicit information held by the knowledge owner (El-Den & Sriratanaviriyakul, 2019). Thus, we hypothesize that:

Hypothesis 1. TA has a significant positive influence on TS.

2.2. TS and GIC

Routine business activities use implicit knowledge, which includes feelings, intuition, insights, and personal skills. According to Chen (2008), a company's intangible assets are all covered in GIC. These include relationships, knowledge, and experience relevant to ecological

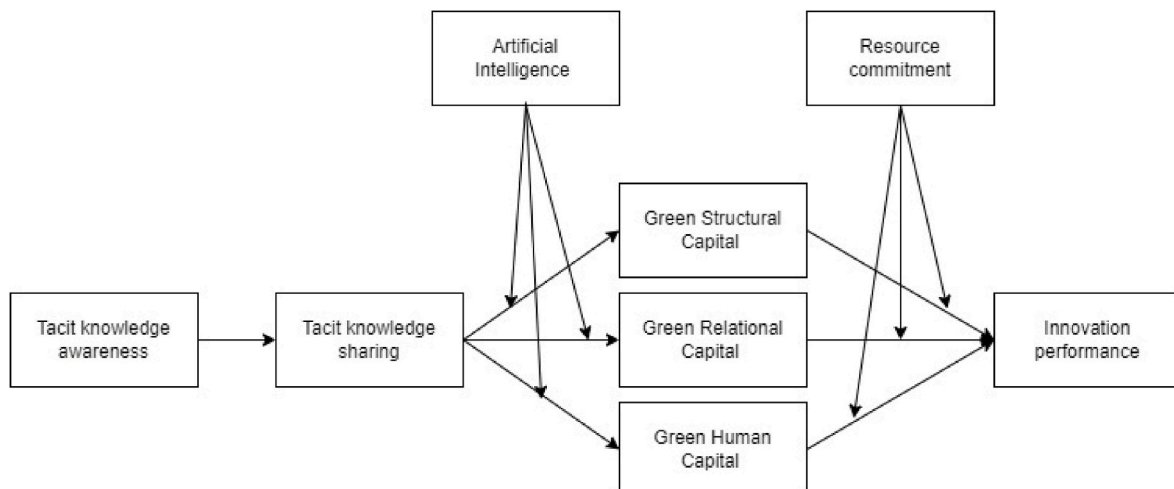


Fig. 1. The conceptual model of the study.

sustainability and green innovation at the person and organizational levels. The three categories of green intellectual capital are GHC, GSC, and GRC. The widespread awareness that knowledge is the foundation of business, and that information is essential for firms led to the founding of GIC. Green structural capital (GSC) includes several aspects like as administrative processes, obligations, knowledge systems, values, workplace culture, reputation, inventions, copyrights, trademarks, etc., that are associated with ecological sustainability or innovation in a firm (Chen, 2008). It is gathered and distributed via organizational frameworks, procedures, systems, and guides (Bontis et al., 2000). Knowledge retention within a company is the concept that knowledge remains inside the organization even when employees go (Youndt & Snell, 2004). The connections between the focal organization and its primary stakeholders, such as customers, sellers, and allies, are known as relationship capital (Subramaniam & Youndt, 2005).

Green human capital (GHC) encompasses an organization's personnel and their attributes, such as expertise, experience, dedication, and drive (Bontis, 1998). Hejase et al. (2016) define GHC as the cumulative sum of workers' understanding, skills, creativity, and involvement in ecological sustainability and innovation. Since human capital is tied to individuals, it leaves the organization when employees depart. Scholars emphasize human capital as the most essential aspect of intellectual capital, as no innovation can occur without it (Spender, 1996). Similarly, green relational capital (GRC) reflects the value derived from external relationships with customers, suppliers, and institutions (Bontis, 1998). GRC refers as the assets related to a firm's interactions with consumers, suppliers, professional connections, and allies about corporate environmental sustainability and ecological innovation (Chen, 2008). Thus, the elements of GIC are influenced by the management of tacit knowledge, which serves as a crucial organizational asset for green knowledge, enabling individuals or organizations to gain competence in environmental sustainability management. Collaborating towards a common goal enhances performance through sharing personal experiences, reflecting, and applying specific skills. Tacit knowledge sharing occurs formally through training or conferences and informally through task groups and social interactions. This sharing reshapes organizational practices, processes, and culture, influencing both cognition and behavior (Caputo et al., 2019a; Holste & Fields, 2010; Shi, 2023).

Tacit knowledge, being inherent in action, situated in practical experience, and influenced by individuals' judgements, is difficult to formalize, recognize, access, and communicate to others (Baronian, 2022). Tacit information sharing occurs in the "place of knowledge making" via the interactions of people, groups, and units, enhancing the knowledge of workers who hold implicit knowledge, ultimately contributing to organizational knowledge production (Nonaka & Von Krogh, 2009). Sharing tacit information enables team members to acquire practical skills, unique experiences, and specialized techniques, enhancing performance through cognitive innovation (Bloodgood & Chilton, 2012). This process improves stakeholders' common sense, social skills, intuition, emotions, and unspoken mental models, contributing to a firm's human capital. As tacit knowledge is a core competitive advantage, recipients value the subjective, context-specific insights provided by sharers. Collaborative programs further promote tacit knowledge sharing, fostering mutual understanding, trust, and cohesiveness among team members (Peng & Lai, 2012). Prior studies have shown that when tacit knowledge sharing takes place, both parties refrain from making demands that could harm mutual interests and instead strengthen their relationship by engaging in activities like sharing preferences, exchanging personal experiences, and demonstrating goodwill and care (Hsu & Wang, 2012). Thus, we hypothesize.

Hypothesis 2. TS has a positive influence on the a) GSC, b) GHC, and c) GRC.

2.3. Influence of GIC on innovative performance (IP)

GSC greatly enhances a company's ability to be innovative, giving it a robust method for handling and sharing knowledge within the company (Khan et al., 2021). The organization's structure and routines give companies a steady way to use knowledge, building and updating their offerings and innovations. By improving organizational learning capability, green structural capital helps to improve innovation and sustain competitive advantage (Caputo et al., 2019b; Chaudhary et al., 2023). The creation of new products, processes or activities needs to deal with various types of knowledge and the institutionalized wisdom and codified experience will provide the innovation impetus (Dhir et al., 2023). Because of organizational memory, relevant ideas can be easily accessed and combined to encourage the company's ongoing progress (Fleming & Sorenson, 2004; Walsh & Ungson, 1991). An organizations' established norms, structures, and routines support a systematic retention and documentation of knowledge, making organizations to endlessly produce and test novel ideas (Hargadon & Sutton, 2000). Information and communication technologies support innovation by making it possible for organizations to search for, retrieve, store, send, analyze and disseminate information (Alavi & Leidner, 2001). To conclude, we can infer that structural capital is conducive to innovation as it leads to a common infrastructure between the act of growing knowledge-creation activities within the firm. Organizational structure, culture, rules, and guidelines all support innovation-related competencies that enhance innovation activities (Aramburu & Sáenz, 2011). In a similar vein, businesses create structural capital to enhance organizational learning capacity (Ulrich et al., 2006), which boosts their performance in terms of innovation (Hsu & Wang, 2012).

Hypothesis 3. GSC is positively associated with IP.

GHC plays a pivotal role in driving innovation performance by equipping employees with the environmental expertise necessary to develop sustainable, innovative products and processes (Bhatt et al., 2020). As firms invest in GHC, they unlock their potential to enhance green innovation and achieve long-term competitive advantage through environmentally conscious strategies. Subramaniam and Youndt (2005) assert that a significant amount of the knowledge and skills required for innovation reside within individuals and are actively applied by them. Similarly, De Winne and Sels (2010) emphasize that acquiring prior knowledge is essential for developing new insights. Employees' expertise drives organizational innovation, as creative and knowledgeable individuals generate innovative ideas (Anand et al., 2007) and challenge established norms, acting as catalysts for organizational change (Amabile, 1997). Human capital, therefore, is a vital resource for innovation, especially in today's fast-evolving technological landscape (Subramaniam & Youndt, 2005). For instance, employees with environmental expertise are more likely to be motivated to apply this knowledge to sustainable innovations. Greater investment in GHC can significantly boost green innovation, as organizations leverage their GHC to develop environmentally friendly products and processes. GHC serves as a bridge, linking employee environmental knowledge to green innovation efforts, enhancing both innovation performance and organizational sustainability. However, employees who lack awareness of environmental issues may struggle to adopt the necessary attitudes and behaviors to address such challenges (Wang & Juo, 2021).

Hypothesis 4. GHC is positively associated with IP.

GRC plays a crucial role in driving innovative performance by fostering strong external partnerships that facilitate the exchange of environmental knowledge and ideas (Talwar et al., 2021). Through collaboration with environmentally conscious stakeholders, firms can co-create sustainable innovations, enhancing both green product development and eco-friendly processes. GRC is closely linked to creativity and innovation, as not all knowledge necessary for innovation resides within a firm's boundaries. According to Kale et al. (2000),

organizations can benefit from external relationships by accessing existing solutions or integrating diverse external knowledge sources into their innovation processes. Relational capital not only enhances power, control, and influence, but also fosters mutual solidarity, increasing the likelihood of co-creation. The involvement of external stakeholders, such as customers, in development activities further stimulates continuous innovation (Shaik et al., 2025). Pérez-Luño et al. (2011) suggest that inter-organizational cooperation enhances innovation by facilitating knowledge exchange and interactive learning. For instance, companies need to embed state of the art environmental expertise to form strategic alliances with stakeholders, as such collaborations support better innovation outcomes and encourage collective invention. By leveraging GRC in a cooperative environment, companies have opportunities to advance new environmental innovations and ideas (Hart, 1995; Saviano et al., 2018). Sustainable business relationships can provide incentives to companies to create green products and to innovate environmentally friendly processes to meet the needs of environmentally oriented customers. Thus, GRC-driven collaboration is necessary to ensure sustainable innovation and maximized time performance of an organization.

Hypothesis 5. GRC is positively associated with IP.

2.4. Role of artificial intelligence (AI) as a moderator

AI plays a transformative role in enhancing GSC by enabling organizations to efficiently manage, store, and retrieve institutional knowledge (Dhir et al., 2024; Saviano et al., 2023), thereby boosting their innovation capabilities. Through AI, GHC is empowered, as employees can access advanced tools for skill development and knowledge sharing, fostering creativity and innovation. Furthermore, AI strengthens GRC by facilitating seamless communication and collaboration across networks, allowing businesses to leverage external knowledge for continuous innovation (Madanaguli et al., 2023). Raman et al. (2025) claim that AI is an emerging field that is used in all industries to raise efficiency and productivity. AI is a tool that employs human intellect to increase performance in multiple domains. AI is one of the most significant technical developments of the twenty-first century. It has already begun to change societies and commercial organizations in ways that were unimaginable only a few decades ago. According to Chowdhury et al., (2022), AI is becoming an essential part of business innovation initiatives, process modernization, disruption, and establishing a competitive edge in companies that have embraced a data-driven and digital environment. Even while it is impossible to ignore in the current digital era, artificial intelligence plays a crucial role in organizational performance and sustainability, this role has received less attention. Within the topic of HR Technology, artificial intelligence is a new frontier that has the potential to either supplement or improve green human resource (Garg et al., 2018). Applying AI is one way to deliver knowledge that is ready for use in making decisions by employees. Moreover, employees prefer using AI technologies to help in filling the knowledge gap that adds value to bottom-line activities (Mazuze, 2019). Under the influence of AI, GRC strongly affects the organization's willingness to implement green management practices. The significance of AI in the workplace has been demonstrated by a number of scientific studies (Li et al., 2019).

AI serves as a pivotal moderator in the intricate interplay between TKS and GSC, orchestrating the transformation of unstructured, intangible knowledge into actionable and systematized insights that drive organizational sustainability and innovation. In manufacturing SMEs in the UAE, AI-based knowledge management solutions such as SAP's sustainability management solutions support the institutionalization of sustainable practices. These platforms store, retrieve best practices and speed-up the green innovation adoption (Lu et al., 2025b; Chowdhury et al., 2022). For example, AI technology makes it easier to capture tacit knowledge on energy-efficient activities, making that knowledge accessible to all and support efforts related to innovation. Furthermore,

UAE SMEs also employ AI based enterprise resource planning (ERPs) to standardize and spread the sustainable manufacturing practices that increase the output efficiency and green compliance (Raman et al., 2025).

AI also significantly improves GHC by empowering customized training and targeted program for skill development. Adaptive AI-based learning platforms help employees to find relevant information about how to perform sustainable manufacturing activities. Canadian SMEs make use of platforms like Coursera for Business to motivate their workforce to learn eco-friendly practices (Anani, 2018), seamlessly assimilating tacit knowledge into daily operational routines. These platforms/channels not only communicate essential facts but embolden individuals to play an active part in green innovation, elevating the role of individuals as agents of sustainability. Additionally, AI-based platforms, like Talent LMS, facilitate this process by developing tailored learning paths for employees, providing instant access to eco-innovation materials, and incorporating green skills throughout business operations (Rodríguez et al., 2024). In addition, AI brings GRC benefits by encouraging open and trusting partnerships between firms and their outside stakeholders (Zhou et al., 2024). Likewise, SMEs use predictive analytics and AI-powered technology to build stronger links with partners that prioritize the environment. For instance, UAE manufacturing SMEs leverage AI-driven platforms to analyze supplier data, identifying sustainable collaborators and fostering effective green initiatives. Likewise, Canadian SMEs utilize AI tools to cultivate inter-organizational partnerships, advancing collaborative networks that underpin sustainable packaging innovations (Dhar, 2024). For example, tools such as AI-enhanced Slack channels further facilitate interactive learning between internal teams and external partners, enabling efficient sharing of sustainability-focused insights while nurturing trust and cohesion in eco-partnerships (Peng & Lai, 2012). Based on the above discussion, therefore, we propose that AI moderates the relationship between TKS and GSC, GHC, and GRC by improving the accessibility, scalability, and integration of tacit knowledge across organizational processes and stakeholder networks. This transformative role of AI facilitates the institutionalization of sustainable practices (Ghobakhloo et al., 2023), while simultaneously fostering innovation enhancing collaboration, and supporting skill development.

Hypothesis 6. AI moderates the relationship between TS and a) GSC, b), GHC and c) GRC

2.5. Role of resource commitment (RC) as a moderator

Implementing technological breakthroughs requires complex and dynamic changes, necessitating a supportive internal environment within the company (Caputo et al., 2021; Koc & Ceylan, 2007). To foster innovation, organizations must ensure that employees are both capable (i.e., provided with opportunities) and motivated to innovate. The primary goal of management should be to cultivate and sustain a creative internal culture (Prajogo & Ahmed, 2006). Establishing such a culture requires a firm commitment to innovation, as it fosters the development and integration of knowledge necessary for effective innovation. RC refers to the allocation of an organization's current and future resources. Berrone et al. (2013) highlight that a firm's resource profile fundamentally constrains and shapes its strategy, meaning that the decision of how much to invest in innovation must be carefully considered. When focusing on eco-innovation, businesses prioritizing sustainability typically need to allocate more resources and face longer time frames to recover their investments (Ambec & Lanoie, 2008). This is because eco-innovation often involves greater novelty and uncertainty compared to traditional innovation (Cainelli et al., 2015; Jabeen et al., 2023; Lu et al., 2025a). Environmentally conscious businesses can achieve greater success by directing resources toward exploring new green knowledge, technologies, and business models aligned with creative outputs and emerging markets (Richey & Klein, 2014). Green-oriented SMEs can

effectively reduce innovation costs by leveraging resource commitment to select optimal resource combinations (Katsikeas et al., 2016). RC thus provides eco-conscious firms with the flexibility to pursue long-term, strategic eco-innovation goals (Berrone et al., 2013). While RC significantly impacts organizational innovation, its effect on individual employees' innovative performance remains underexplored. It could potentially enhance or weaken the direct relationship between GIC and innovative performance. Therefore, we propose the following hypothesis.

Hypothesis 7. RC moderates the relationship between a) GSC, b) GHC, c) GRC and Innovation performance.

2.6. Mediating role of GIC

Businesses seek effective AI solutions to allocate resources efficiently, deliver premium goods and services, enhance organizational effectiveness, and create new business opportunities (Shah et al., 2023). Their expertise, access to big data, and information-driven strategies contribute to increased creativity and success. Relationship capital, particularly in the context of sustainability and innovation, has expanded multidisciplinary discussions initiated by human, social, and structural capital (Ker et al., 2014). Trust-based interorganizational relationships further foster creativity and knowledge exchange (Sankowska, 2013). As Furr et al. (2016) note, businesses often partner to grow their ecosystem, strengthen their competitive edge, and unlock creative potential. Organizational, financial, and human capital are critical factors influencing business processes and value creation, particularly in SMEs (Dameri & Ferrando, 2021). Since tacit knowledge is embedded in employee behavior, it is learned through contextual human interactions (Mansingh et al., 2009). Internal relational ties facilitate the conversion of data into knowledge flows (Dobrzykowski & Tarafdar, 2015), while communication networks help employees remain adaptable during crises (Sanford et al., 2020). Despite its importance, tacit knowledge remains difficult to recognize, extract, and convey due to its embedded, context-dependent nature.

KS fosters the expansion of organizational knowledge by enhancing employees' "knowing more than we can tell" through interactions among individuals, teams, and departments (Nonaka & Von Krogh, 2009). This sharing of implicit information allows team members to learn and apply unique experiences, practical skills, and creative methods, ultimately improving their cognitive innovation performance (Bloodgood & Chilton, 2012). Tacit knowledge strengthens social skills, common sense, tactile perception, intuition, emotions, and stakeholders' mental models—key elements of human capital in any firm. In today's dynamic and ever-changing economy, human capital is seen as a critical strategic asset for maintaining a competitive advantage. Organizations require knowledgeable decision-makers, reliable employees, and proficient problem solvers (Campbell et al., 2012). Human capital underpins a firm's capacity to innovate, enhance operational performance, and set new efficiency and quality standards (Cabello-Medina et al., 2011). Companies that prioritize investing in their workforce tend to have more inventive employees, as human capital is the cornerstone of innovation.

KS is demonstrated through attitudes, instincts, perceptions, and individual abilities evident in everyday organizational activities. When individuals share ideas, experiences, and personal insights toward a common goal, collaboration becomes more effective. According to Holste and Fields (2010), some forms of tacit knowledge are formal, acquired through conferences or training, while others are informal, gained through interdepartmental teams, social networks, and employee relationships. The creation of tacit knowledge not only shifts individual attitudes and behaviors but also introduces new ideas into organizational practices, education systems, and broader societal customs, potentially transforming companies. Through collaborative projects, the interactional dynamics of tacit knowledge sharing promote mutual

understanding, trust, and cohesiveness (Peng & Lai, 2012). For instance, parties engaged in tacit knowledge sharing tend to seek common ground, exchange personal experiences, and show empathy, which strengthens relationships and avoids actions that could damage shared interests. Similarly, as Ordóñez de Pablos (2004) suggested, companies enhance structural capital by improving problem-solving, reducing costs, increasing quality, promoting technical exchange, and strengthening processes. This integration of structural capital into business strategy leads to affordable, high-quality products and innovative knowledge dissemination (Aramburu & Sáenz, 2011). Hence, we propose the following indirect hypothesis.

Hypothesis 8. GSC mediates the relationship between TS and IP

Hypothesis 9. GHC mediates the relationship between TS and IP.

Hypothesis 10. GRC mediates the relationship between TS and IP.

3. Methods

3.1. Data collection procedure

Manufacturing firms are crucial drivers of economic development, innovation, and employment in both emerging and developed economies (Ndubis et al., 2021). Their adaptive nature and flexibility make them a prime focus for empirical study of the integration of tacit knowledge, AI, and GIC in fostering innovation. The manufacturing industry operates on a highly knowledge-intensive mechanism involving a blend of explicit and tacit knowledge for improving processes, creating innovative products, and maintaining sustainable practices. SMEs operating in the industry experience peculiar challenges common to their line of business, including limited resources, skill shortages, and regulatory pressures, thus providing a rich context for investigating the ways in which the RC and AI can boost the development of GIC and innovation performance. By examining manufacturing SMEs, this study fulfills an important research gap in investigating strategies that will keep firms competitive in a rapidly changing technological and environmental context. The exclusion of industries other than manufacturing is mainly justified on grounds of research focus on investigating the dynamics between tacit knowledge, AI, and GIC. Industrial sectors present considerable differences in their functional structure, regulatory regimes, and dependency on tacit knowledge, which may introduce confounding variables and diminish the specificity of the research findings. For example, sectors dealing with less tangible outputs may be more dependent on explicit knowledge systems, and hence the effect of tacit knowledge and the interaction of tacit and explicit knowledge with AI may be less significant or not as crucial. Additionally, certain industries operate under vastly different environmental and technological constraints, leading to inconsistencies in the applicability of the study's theoretical model. Focusing exclusively on the manufacturing sector ensures an analysis of its knowledge-intensive and process-driven nature, aligning with the study's objectives.

A cohort of rigorously trained business graduate students conducted the data collection process, maintaining the study's professionalism and integrity. Prior to initiating data collection, the team meticulously reviewed the research timeline and site characteristics, ensuring participants met specific, predefined criteria representative of the target population (Mehmood et al., 2025c). Only individuals directly relevant to the research objectives were included. Senior-level managers in manufacturing SMEs were initially briefed on the research goals and significance to gauge their willingness to participate, and contact details were collected from those who consented. A time-lagged methodology was utilized in order to overcome temporal complexities, and each respondent received a randomly generated unique numbering on pre-stamped emailed questionnaires in order to facilitate accurate matching of answers at time intervals. Pilot testing on a sample of ten respondents provided minor refinement of scale items and resulted in a

refined questionnaire for better clarity and contextual relevance to SMEs in both the UAE and Canada.

A three-phase data collection technique was used to control for social desirability and common method variance, and this technique focused on collecting data over a period of three intervening monthly timeframes. In accordance with the suggestions presented by Podsakoff et al. (2024) and Mehmood et al. (2024a), the time-lagged design of this survey allowed data to be gathered from different sources and time and reflectivity of the responses collected from the respondents, thereby reducing common source and common method biases. Subjects were informed their responses would be used for research purposes and about the confidentiality and non-identifiability of their responses as well as that there were no “right” or “wrong” answers. In line with recent empirical research (e.g., Al Bloushi et al., 2024; Mehmood et al., 2025b), this method not only enhanced the quality of data but also ensured flexibility in aligning the research goals with the specific temporal requirements of the study. By capturing data in three waves after a spacing of a month each time, this robust method reduced the impact of biases and enhanced the validity of the study (Alkhatheeri et al., 2021).

For empirical validation of the model, two culturally distinct contexts, the UAE and Canada, were selected. We believe that the inclusion of variables such as tacit knowledge sharing and awareness, as well as green intellectual capital, within these culturally diverse settings will provide deeper insights. We aimed to test the proposed model (Fig. 1) by using a time lagged research design and collected data in the UAE across multiple waves, focusing on top management of small and medium-sized firms in the manufacturing industry. This demographic was selected because these participants have the experience and insight necessary to assess the opportunities and challenges related to GIC and innovation. We focused on SMEs with at least 100 employees, assuming that these firms have a more formalized management structures and innovation system. The data collection process unfolded in three distinct phases. Initially, at Time Wave 1, we gathered information on variables such as tacit knowledge awareness, tacit knowledge sharing, artificial intelligence, resource commitment, alongside demographic details of the respondents (gender, age, and firm size). Subsequently, one month later at Time Wave 2, we assessed the green intellectual capital. Another month passed before the third wave (Time Wave 3), during which we evaluated the innovation performance. This staggered approach was strategic, aimed at reducing common method bias (CMB) and establishing a clear temporal relationship between the variables, echoing the methodology advocated by Podsakoff et al. (2003). The UAE was selected as the context for this investigation due to its manufacturing sector’s significant contribution to the national GDP-14 % in 2015, with ambitions to increase this figure to 20 % by 2021 and 25 % by 2025. SMEs dominate this sector, with Alefari et al. (2020) reporting that they constitute 94 % of the manufacturing companies. Participation was voluntary, with an incentive of a 100AED prize for those completing all survey phases, and assurances of data confidentiality and academic usage were provided. The data collection commenced with the distribution of 500 surveys in the first wave, yielding 473 completed responses. The second wave saw these respondents contacted again, resulting in 419 returns. The final wave, conducted a month later, led to 317 completed surveys from the same cohort, indicating a 63.4 % response rate, and forming the basis for subsequent statistical analysis.

Within the scope of this study, we employed a time-lagged approach, conducting research across multiple temporal phases to collect data from Canadian manufacturing SMEs. Specifically, the study targeted SMEs that are ISO 14001 certified, focusing on garnering insights through survey-based questionnaires addressed to senior-level management within these organizations. The Canadian manufacturing sector constitutes a pivotal component of the nation’s economic expansion. In 2021, the sector witnessed a significant revenue increase, totaling \$787.3 billion, which marks a \$108.9 billion rise (+16.1 %) compared to 2020 and exceeds the 2019 figures by \$39.1 billion, as reported by Statistics Canada. The initial wave of data collection saw the

distribution of 500 questionnaires, from which 449 valid responses were received, reflecting a diverse range of perspectives from senior managers within the manufacturing SMEs. These responses, collected during the first survey wave, included demographic details and addressed variables such as tacit knowledge awareness, tacit knowledge sharing, artificial intelligence, and resource commitment. Subsequent to this, the second survey wave, conducted a month later, aimed at capturing insights on green intellectual capital from the same 449 participants, yielding 403 useable responses. The final wave focused on innovation performance, from which 383 valid responses were obtained, indicating a 76.6 % response rate. The distribution and demographic details of the UAE and Canadian sample respondents are systematically presented in Table 1.

3.2. Measurements

In this research, all constructs were derived from established scales documented in existing literature, ensuring a robust foundation for the study’s methodology. Given that study targeted participants from the UAE, where English is not the primary language, it necessitated the translation of the original English questionnaires into Arabic. The translation process followed the criteria of back-translation suggested by Brislin (1980) in order to ensure the equivalence of the original survey items. On the other hand, the Canadian sample study used the original English language scales without any change. The tacit knowledge awareness and tacit knowledge sharing were measured using a 4-item scale each developed by Kucharska (2022). Artificial Intelligence was also measured on a 5-item scale established by Dubey et al. (2020), and resource investment was measured using a 4-item scale developed by Konadu et al. (2020). The dimensions of GIC, namely green human capital, green structural capital, and green relational capital, were measured following the adapted instruments of Buenechea-Elberdin et al. (2018) and later used by Shehzad et al. (2023), each comprising 3, 4, and 4 items. Innovation performance was measured using a 5-item scale, adopted from Kianto et al. (2017). All responses were collected on a 5-point Likert scale, “1 (strongly disagree) to 5 (strongly agree)” to allow for detailed feedback from participants. Furthermore, demographic information (i.e., gender, age, and firm size) was also collected as controlled variables in the study, to have a more thorough perspective of the influence of the variables in the context of the investigation. More detailed information on these constructs (i.e., their factor loadings, reliabilities, and validities) is presented in Table 2.

3.3. Common methods bias (CMB)

We implemented both procedural and statistical remedies to address CMB. Procedurally, we mitigated the risk of socially desirable responses by guaranteeing respondent anonymity, as outlined in the survey’s cover letter, following Podsakoff et al. (2024), Khan et al. (2023), and

Table 1
Demographics information.

Demographics	Canada (383)		UAE (317)	
	Frequency	%	Frequency	%
Gender				
Female	168	43.9	143	45.1
Male	215	56.1	174	54.9
Age (in years)				
20 to 25	56	14.6	46	14.5
26 to 30	157	41.0	132	41.6
31 to 35	108	28.2	81	25.6
36 and above	62	16.2	58	18.3
Firm Size (no of employees)				
Less than 100	53	13.8	49	15.5
101 to 200	136	35.5	113	35.6
201 to 300	95	24.8	76	24.0
301 to 400	54	14.1	42	13.2
above 400	45	11.7	37	11.7

Table 2
Studied variables loadings.

Constructs	Coding	Factor loadings		CRs		AVE	
		UAE	CAN	UAE	CAN	UAE	CAN
Tacit Knowledge awareness (TA)				.870	.862	.626	.611
	TA1	.736	.712				
	TA2	.835	.821				
	TA3	.800	.796				
	TA4	.790	.792				
Tacit knowledge sharing (TS)				.864	.857	.616	.602
	TS1	.752	.770				
	TS2	.694	.669				
	TS3	.886	.857				
	TS4	.794	.796				
Artificial Intelligence (AI)				.901	.901	.648	.646
	AI1	.757	.734				
	AI2	.739	.733				
	AI3	.861	.878				
	AI4	.870	.883				
	AI5	.787	.778				
Green Structural Capital (GSC)				.906	.888	.709	.668
	GSC1	.817	.725				
	GSC2	.942	.914				
	GSC3	.874	.881				
	GSC4	.717	.731				
Green Human Capital (GHC)				.888	.880	.725	.710
	GHC1	.879	.879				
	GHC2	.883	.860				
	GHC3	.789	.785				
Green Relational Capital (GRC)				.802	.803	.506	.508
	GRC1	.821	.839				
	GRC2	.682	.694				
	GRC3	.648	.641				
	GRC4	.682	.660				
Resource Commitment (RC)				.916	.909	.735	.718
	RC1	.722	.708				
	RC2	.774	.744				
	RC3	.959	.954				
	RC4	.950	.953				
Innovation performance (IP)				.934	.926	.739	.714
	IP1	.881	.855				
	IP2	.897	.887				
	IP3	.875	.864				
	IP4	.796	.789				
	IP5	.846	.829				

Notes: UAE = United Arab Emirates, CAN = Canada.

Mehmood et al. (2025a). We also enhanced the survey's item clarity and minimized ambiguity by clearly defining scale items and framing questions to be direct and specific. Measures were taken to avoid linking variables or allowing for educated guesses by refraining from labeling variables by reported constructs and not grouping items by variable, adhering to Mehmood et al. (2025b)'s recommendations. Furthermore, we emphasized the absence of "correct" answers and assured respondents of the confidentiality of their responses. Statistically, we applied a common latent factor (CLF) approach, assessing regression coefficients before and after incorporating CLF into the model, finding no dominant factor indicative of CMB.

Additionally, we adopted a confirmatory factor analysis (CFA) to rigorously assess the potential for common method bias within our study. The analysis was applied separately to data from the UAE and Canada samples. For the UAE sample, the 1-factor model demonstrated a suboptimal fit, evidenced by the following statistical indicators: $\chi^2 = 4894.319$, $df = 495$, $\chi^2/df = 9.887$, CFI = .368, TLI = .326, and RMSEA

= .168. Correspondingly, the Canada sample analysis showed a poor fit of the single-factor model by $\chi^2 = 5585.711$, $df = 495$, $\chi^2/df = 11.284$, CFI = .354, TLI = .311, and RMSEA = .164, which altogether indicated the nonexistence of CMB. The same conclusion was supported by Harman's single-factor test, as it indicated that the variance explained by the principal factor for both the samples was a minor percentage and hence disproved the occurrence of significant CMB in our research work. Similar findings can be seen from Mehmood et al. (2024b), Khan et al. (2024), and Shang et al. (2021), supporting the transparency of our data and the validity of our findings. Moreover, to take into account the multicollinearity issues between the variables, we followed the procedure suggested by Hair et al. (2011) and Mehmood et al. (2024c). The VIF values were below 5 for the datasets in both samples. This implies multicollinearity is not a serious problem in the present study.

3.4. Reliability and validity

We conducted extensive analyses to test the reliability and validity of our dataset; using Cronbach's alpha (CA), composite reliability (CR), and average variance extracted (AVE) as the main metrics. The results, shown in Table 2, revealed that the CA, CR and AVE of both of the studies were above the minimum recommended level of .70, .80, and .50, respectively, as suggested by Cheng et al. (2024), Yu et al. (2021), and Hair et al. (2003). This confirms the validity and convergent validity of our data in both samples. Additionally, the convergent validity of our research model was further evaluated through the analysis of factor loadings for each item on their designated variables, with Table 2 showcasing that all factor loadings exceeded the benchmark value of .60 for both samples. The correlation matrix for the study's constructs is detailed in Table 3. For assessing discriminant validity, we compared the square root values of AVE for all constructs against their correlation coefficients, finding the former to be consistently higher, thereby affirming the model's discriminant validity.

4. Results

4.1. Correlation and CFA

Table 3 presents the correlation matrix and descriptive statistics for the UAE and Canadian samples, revealing significant correlations that preliminarily support our theoretical model's validity. We conducted CFA by using AMOS to assess the distinctiveness of variables such as TA, TS, AI, GSC, GHC, GRC, RC, and IP (see Table 4). For the UAE sample, the 8-factor model demonstrated superior fit ($\chi^2/df = 720.388/467 = 1.543$; CFI = .964; TLI = .959; RMSEA = .041). These included a 7-factor model combining TA, TS, AI, GSC, GHC, GRC, and IP ($\chi^2/df = 1329.113/474 = 2.804$; CFI = .863; TLI = .861; and RMSEA = .076), a 6-factor model combining TA, TS, GSC, GHC, GRC, and IP ($\chi^2/df = 1823.345/480 = 3.799$; CFI = .807; TLI = .788; and RMSEA = .094), a 5-factor combining GSC, GHC, GRC, RC, and IP ($\chi^2/df = 2498.154/485 = 5.151$; CFI = .711; TLI = .685; and RMSEA = .115), a 4-factor combining GSC, GHC, GRC, and IP ($\chi^2/df = 2795.863/489 = 5.718$; CFI = .669; TLI = .642; and RMSEA = .122), a three factor combining GSC, GHC, and GRC ($\chi^2/df = 2980.649/492 = 6.058$; CFI = .643; TLI = .617; and RMSEA = .127), a 2-factor combining TA and IP ($\chi^2/df = 3896.915/494 = 7.888$; CFI = .511; TLI = .478; and RMSEA = .148) and the 1-factor model showing a poor fit ($\chi^2/df = 4894.319/495 = 9.887$; CFI = .368; TLI = .326; and RMSEA = .168).

The analysis of the Canada sample similarly established the 8-factor model's superior fit ($\chi^2/df = 728.045/467 = 1.559$; CFI = .967; TLI = .963; RMSEA = .038), surpassing the fit of alternative models such as a 7-factor model combining TA, TS, AI, GSC, GHC, GRC, and IP ($\chi^2/df = 1425.565/474 = 3.008$; CFI = .879; TLI = .865; and RMSEA = .072), a 6-factor model combining TA, TS, GSC, GHC, GRC, and IP ($\chi^2/df = 2015.635/480 = 4.199$; CFI = .805; TLI = .786; and RMSEA = .092), a 5-factor combining GSC, GHC, GRC, RC, and IP ($\chi^2/df = 2826.918/485$

Table 3
Descriptive statistics.

	1	2	3	4	5	6	7	8
UAE Mean	2.85	3.95	3.85	3.65	3.90	3.73	3.93	3.82
UAE SD	1.04	.92	1.01	1.13	1.06	1.02	.92	1.12
1. Tacit Knowledge awareness		.273**	.355**	.264**	.221**	.285**	.201**	.218**
2. Tacit knowledge sharing	.237**		.240**	.325**	.446**	.417**	.265**	.260**
3. Artificial Intelligence	.332**	.240**		.394**	.331**	.355**	.262**	.252**
4. Green Structural Capital	.244**	.294**	.334**		.588**	.429**	.251**	.350**
5. Green Human capital	.224**	.406**	.286**	.604**		.506**	.225**	.325**
6. Green relational capital	.282**	.385**	.304**	.395**	.471**		.232**	.237**
7. Resource Commitment	.149**	.236**	.186**	.219**	.209**	.195**		.320**
8. Innovation performance	.190**	.238**	.181**	.369**	.343**	.249**	.276**	
Canada Mean	2.87	3.96	3.88	3.73	3.97	3.77	3.93	3.87
Canada SD	1.03	.91	1.03	1.09	1.02	1.01	.91	1.08

Note: Upper triangle = UAE; lower triangle = Canada, **p < .01, *p < .05.

Table 4
CFA.

Models	χ^2	df	χ^2/df	$\Delta \chi^2 (\Delta df)$	TLI	CFI	RMSEA
UAE Sample							
Hypothesized 8-Factor Model ^a	720.388	467	1.543	–	.959	.964	.041
7-Factor Model ^b	1329.113	474	2.804	608.725(7)	.861	.863	.076
6-Factor Model ^c	1823.345	480	3.799	494.232(6)	.788	.807	.094
5-Factor Model ^d	2498.154	485	5.151	674.809(5)	.685	.711	.115
4-Factor Model ^e	2795.863	489	5.718	297.709(4)	.642	.669	.122
3-Factor Model ^f	2980.649	492	6.058	184.786(3)	.617	.643	.127
2-Factor Model ^g	3896.915	494	7.888	916.266(2)	.478	.511	.148
1-Factor Model ^h	4894.319	495	9.887	997.404(1)	.326	.368	.168
Canada sample							
Hypothesized 8-Factor Model ^a	728.045	467	1.559	–	.963	.967	.038
7-Factor Model ^b	1425.565	474	3.008	697.520(7)	.865	.879	.072
6-Factor Model ^c	2015.635	480	4.199	590.070(6)	.786	.805	.092
5-Factor Model ^d	2826.918	485	5.829	811.283(5)	.676	.703	.112
4-Factor Model ^e	3161.059	489	6.464	334.141(4)	.634	.661	.120
3-Factor Model ^f	3409.697	492	6.930	248.638(3)	.603	.630	.125
2-Factor Model ^g	4499.225	494	9.108	1089.528(2)	.457	.492	.146
1-Factor Model ^h	5585.711	495	11.284	1086.486(1)	.311	.354	.164

= 5.829; CFI = .703; TLI = .676; and RMSEA = .112), a 4-factor combining GSC, GHC, GRC, and IP ($\chi^2/df = 3161.059/489 = 4.464$; CFI = .661; TLI = .634; and RMSEA = .120), a three-factor combining GSC, GHC, and GRC ($\chi^2/df = 3409.697/492 = 6.930$; CFI = .630; TLI = .603; and RMSEA = .125), a 2-factor combining TA and IP ($\chi^2/df = 4499.225/494 = 9.108$; CFI = .492; TLI = .457; and RMSEA = .146) and lastly a 1-factor model showing a poor fit ($\chi^2/df = 5585.711/495 = 11.284$; CFI = .354; TLI = .311; and RMSEA = .164). Convergent validity was confirmed with significant loadings ($p < .01$) for all observed items, validating discriminant validity and affirming the model's robustness for further hypothesis testing.

4.2. Hypothesis testing

Following the methodology recommended by Hayes (2018), this study's proposed model was analyzed using PROCESS macros within SPSS. The analysis indicated that the controlled variables exerted no significant influence on the primary constructs of this study across both the UAE and Canadian samples. Specifically, Tables 5 and 6 present the empirical findings from the UAE and Canadian contexts, respectively offering substantial validation for our research hypotheses. In particular, our analysis demonstrated a positive association between TA and TS in both samples, with the UAE sample showing a [unstandardized $\beta = .242$, SE = .048, 95 % CI [.148, .340]] and the Canadian sample mirroring this result (unstandardized $\beta = .209$, SE = .044, 95 % CI [.123, .293]), thereby affirming H1 in both samples.

Further, Table 5 reveals a significant positive relationship between TS and GSC (unstandardized $\beta = .399$, SE = .065, 95 % CI [.247, .540]), TS and GHC (unstandardized $\beta = .516$, SE = .058, 95 % CI [.364, .651]),

Table 5
Hypotheses results.

Hypothesis (UAE sample)	β	S.E.	Boot LL 95 % CI	Boot UL 95 % CI
H1: TA → TS	.242	.048	.148	.340
H2a: TS → GSC	.399	.065	.247	.540
H2b: TS → GHC	.516	.058	.364	.651
H2c: TS → GRC	.459	.056	.325	.577
H3: GSC → IP	.295	.055	.187	.403
H4: GHC → IP	.276	.063	.153	.399
H5: GRC → IP	.172	.066	.043	.302
Moderating effects				
H6a: AI * TS → GSC	.200	.052	.098	.303
H6b: AI * TS → GHC	.175	.048	.082	.269
H6c: AI * TS → GRC	.188	.045	.099	.278
H7a: RC * GSC → IP	.192	.052	.089	.294
H7b: RC * GHC → IP	.211	.049	.114	.308
H7c: RC * GRC → IP	.227	.058	.113	.340

Note: Tacit Knowledge awareness = TA, Tacit knowledge sharing = TS, Artificial Intelligence = AI, Green Structural Capital = GSC, Green Human capital = GHC, Green relational capital = GRC, Resource Commitment = RC, Innovation performance;

and TS and GRC (unstandardized $\beta = .459$, SE = .056, 95 % CI [.325, .577]) within the UAE sample. Consequently, Hypotheses 2a, 2b, and 2c are substantiated in the UAE context. Similarly, the Canadian sample exhibits statistically significant positive relationships between TS and GSC (unstandardized $\beta = .352$, SE = .059, 95 % CI [.210, .484]), TS and GHC (unstandardized $\beta = .457$, SE = .053, 95 % CI [.320, .583]), and TS and GRC (unstandardized $\beta = .429$, SE = .053, 95 % CI [.304, .547]), thus supporting Hypotheses 2a, 2b, and 2c in Canada as well.

Table 6
Hypotheses results.

Hypothesis (Canada sample)	Effect	S.E.	Boot LL 95 % CI	Boot UL 95 % CI
H1: TA → TS	.209	.044	.123	.293
H2a: TS → GSC	.352	.059	.210	.484
H2b: TS → GHC	.457	.053	.320	.583
H2c: TS → GRC	.429	.053	.304	.547
H3: GSC → IP	.327	.049	.230	.424
H4: GHC → IP	.313	.056	.204	.422
H5: GRC → IP	.199	.057	.086	.311
Moderating effects				
H6a: AI * TS → GSC	.187	.047	.095	.279
H6b: AI * TS → GHC	.152	.043	.067	.236
H6c: AI * TS → GRC	.178	.042	.095	.261
H7a: RC * GSC → IP	.155	.048	.061	.249
H7b: RC * GHC → IP	.180	.046	.089	.270
H7c: RC * GRC → IP	.149	.052	.046	.252

Note: Tacit Knowledge awareness = TA, Tacit knowledge sharing = TS, Artificial Intelligence = AI, Green Structural Capital = GSC, Green Human capital = GHC, Green relational capital = GRC, Resource Commitment = RC, Innovation performance.

Moreover, the results indicate a significant and positive relation between GSC and IP (unstandardized $\beta = .295$, SE = .055, 95 % CI [.187, .403]), GHC and IP (unstandardized $\beta = .276$, SE = .063, 95 % CI [.153, .399]), and GRC and IP (unstandardized $\beta = .172$, SE = .066, 95 % CI [.043, .302]) in the UAE sample, endorsing Hypotheses 3, 4, and 5 within the UAE context. In the Canadian sample, a similar pattern emerges, with GSC showing a significant positive relationship with IP (unstandardized $\beta = .327$, SE = .049, 95 % CI [.230, .424]), GHC with IP (unstandardized $\beta = .313$, SE = .056, 95 % CI [.204, .422]), and GRC with IP (unstandardized $\beta = .199$, SE = .057, 95 % CI [.086, .311]), thus affirming Hypotheses 3, 4, and 5 in Canada. These findings from both samples collectively provide a robust empirical foundation supporting the validity of the proposed hypotheses, with each hypothesis receiving statistical corroboration.

4.3. Analyzing the moderating role of AI

Table 5 elucidates the moderating effects of AI within the UAE sample, aligning with Hypothesis 6a, which theorized that AI amplifies the relationship between TS and GSC. This hypothesis was substantiated as the interaction between TS and AI significantly predicted GSC (unstandardized $\beta = .200$, SE = .052, $p < .01$). Similarly, Hypothesis 6b, which posited that AI would strengthen the TS-GHC link, was confirmed through a significant interaction effect on GHC (unstandardized $\beta = .175$, SE = .048, $p < .01$). Furthermore, Hypothesis 6c, suggesting that AI enhances the TS-GRC relationship, received empirical support via a notable interaction effect (unstandardized $\beta = .188$, SE = .045, $p < .01$).

In the context of the Canada sample, detailed in Table 6, the findings corroborate the posited hypotheses as follows: Hypothesis 6a, suggesting AI's enhancement of the TS-GSC linkage, was validated by a significant TS and AI interaction effect on GSC (unstandardized $\beta = .187$, SE = .047, $p < .01$). Hypothesis 6b, which proposed that AI would amplify the TS-GHC connection, was evidenced by a significant interaction effect on GHC (unstandardized $\beta = .152$, SE = .043, $p < .01$). Lastly, Hypothesis 6c, asserting that AI would bolster the TS-GRC association, was supported by a significant TS and AI interaction influencing GRC (unstandardized $\beta = .178$, SE = .042, $p < .01$).

4.4. Analyzing the moderating role of RC

Table 5 presents the findings for the UAE sample, illustrating the moderating effects in alignment with our theoretical propositions. Specifically, Hypothesis 7a posited that RC would enhance the relationship between GSC and IP. This hypothesis was confirmed, as

indicated by a significant interaction effect between GSC and RC on IP (unstandardized $\beta = .192$, SE = .052, $p < .01$). Furthermore, Hypothesis 7b, which suggested that RC would strengthen the GHC-IP association, was substantiated by a significant interaction effect (unstandardized $\beta = .211$, SE = .049, $p < .01$). Lastly, Hypothesis 7c argued for the enhancement of the GRC-IP linkage by RC, which was supported by a notable interaction effect (unstandardized $\beta = .227$, SE = .058, $p < .01$).

Similarly, the results for the Canada sample as detailed in Table 6, the analysis reaffirmed the hypotheses. Hypothesis 7a, regarding the amplification of the GSC-IP relationship by RC, was validated again with a significant interaction effect (unstandardized $\beta = .155$, SE = .048, $p < .01$). Hypothesis 7b, concerning the enhancement of the GHC-IP association by RC, was evidenced by a significant interaction effect (unstandardized $\beta = .180$, SE = .046, $p < .01$). Additionally, Hypothesis 7c, which proposed that RC would bolster the GRC-IP connection, was supported by a significant interaction (unstandardized $\beta = .149$, SE = .052, $p < .01$).

4.5. Analyzing the mediating role of GIC

In the examination of GIC as a mediator, a bootstrapping sampling technique with 10,000 resamples was employed to derive 95 % confidence intervals (CIs), thereby facilitating the assessment of indirect relationships between the independent and dependent variables. The analysis conducted on the UAE sample, as presented in Table 7, substantiates the mediating role of GSC in the relationship between TS and IP, with an indirect effect quantified at .118, 95 % CI [.058, .185]. Similarly, GHC was found to mediate the TS-IP relationship, exhibiting an indirect effect of .143, 95 % CI [.066, .234]. Furthermore, GRC also mediated the TS-IP relationship, with an observed indirect effect of .079, 95 % CI [.008, .152]. The exclusion of zero from the lower and upper limits of the 95 % confidence intervals underscores the significance of mediation, thereby affirming the mediation hypotheses H8, H9, and H10 for the UAE sample.

For the Canadian context, the mediation hypotheses were scrutinized utilizing Hayes' (2018) PROCESS macro, as detailed in Table 7. This approach aimed to elucidate the indirect effects of GSC, GHC, and GRC on the TS → IP relationship. The findings revealed that Hypothesis 8, proposing an indirect effect of GSC through TS on IP, was supported, evidenced by a significant indirect effect [.115, 95 % CI (.062, .177)]. Hypothesis 9, which posited an indirect influence of GHC via TS on IP, received confirmation through a significant indirect effect [.143, 95 % CI (.075, .217)]. Lastly, Hypothesis 10, regarding the indirect effect of GRC through TS on IP, was substantiated by the analysis, indicating a

Table 7
Bootstrap results of indirect effect of GIC.

IV	MV	DV	Effect of IV on M (a)	Effect of M on DV (b)	Indirect effect (a*b)	95 % LLCI	95 % ULCI
<i>UAE sample</i>							
H8: TS → GSC → IP							
TS	GSC	IP	.399	.295	.118	.058	.185
H9: TS → GHC → IP							
TS	GHC	IP	.516	.276	.143	.066	.234
H10: TS → GRC → IP							
TS	GRC	IP	.459	.172	.079	.008	.152
<i>Canada sample</i>							
H8: TS → GSC → IP							
TS	GSC	IP	.352	.327	.115	.062	.177
H9: TS → GHC → IP							
TS	GHC	IP	.457	.313	.143	.075	.217
H10: TS → GRC → IP							
TS	GRC	IP	.429	.199	.085	.029	.145

Notes: Tacit knowledge sharing = TS, Green Structural Capital = GSC, Green Human capital = GHC, Green relational capital = GRC, Innovation performance = IP.

significant relationship [.085, 95 % CI (.029, .145)], thus supporting the hypothesis. These results from both the UAE and Canadian samples provide comprehensive validation of the hypothesized mediating roles of GSC, GRC, and GHC, highlighting the pivotal function of green intellectual capital in enhancing innovative performance through tacit knowledge sharing.

5. Discussion

This study explores how manufacturing SMEs can enhance innovation performance through various factors including TS, GIC (comprising GHC, GSC, and GRC), AI, and RC. We propose that TS influences innovation performance in manufacturing SMEs through the development of GIC. Additionally, we investigate how TS affects GIC in the presence of AI, and how the RC of firms stimulates the relationship between GIC and innovation performance. Using the Knowledge-Based View (KBV) paradigm as the conceptual framework, we analyzed the proposed relationships among these constructs. The research concluded with several important findings and confirmed all the postulated hypotheses. The findings show that TK, TS, GIC, AI and RC all significantly influence the innovation performance of manufacturing SMEs in Canada and in the UAE. Some of the major implications of this study include the validation that TS directly affects GIC, which in turn affects innovation performance. AI were identified as a significant moderator that may influence further the relationship between TS and GIC. In addition, RC proved to be a key element to reinforce the relationship between GIC and innovation performance, evidencing the relevance of the strategic allocation of resources to foster innovativeness. These results emphasize the importance of tacit knowledge and its sharing in the context of innovation. The incorporation of AI and resource commitment also enhances the effect of GIC on innovation performance. This study offers insightful implications for the SMEs in the manufacturing sector by which the SMEs can influence these elements to innovate and at the same time to achieve competitive advantage to compete effectively in the market. Overall, the study identifies the interrelationships among TS, GIC, AI, and RC in promoting innovation performance in manufacturing SMEs and provides a comprehensive model that can be used by practitioners to manage firm's effective innovation strategies.

6. Implications

6.1. Theoretical implications

This study makes several significant contributions to the existing body of literature on innovation and knowledge management. Firstly, it extends the understanding of tacit knowledge sharing, specifically within the framework of the KBV of the firm. By exploring the impact of tacit knowledge sharing on innovation performance through green intellectual capital, this research enriches our comprehension of these dynamics in the context of manufacturing SMEs. Additionally, this study uniquely examines the moderating roles of artificial intelligence and resource commitment in these relationships. This perspective introduces a novel dimension to the literature by highlighting how AI interacts with tacit knowledge sharing to influence green intellectual capital (i.e., GSC, GHC, and GRC). The focus on manufacturing organizations in two culturally distinct developed countries, Canada and the UAE, with different levels of innovation performance, adds a comparative aspect that further deepens the analysis.

Additionally, this study provides a stronger theoretical understanding of the relationships among tacit knowledge sharing, green intellectual capital, artificial intelligence, and resource commitment, offering practical prospects for promoting innovation in the manufacturing industry. In doing so, it delivers valuable insights for manufacturing firms aiming to leverage tacit knowledge to enhance innovation performance through green intellectual capital. Moreover, the study deepens the understanding of resource commitment mechanisms by empirically

demonstrating how these mechanisms facilitate innovation by linking green intellectual capital and innovation performance. This comprehensive approach clearly illustrates how various dimensions of intellectual capital serve as bridges connecting tacit knowledge sharing to innovation performance.

6.2. Practical implications

There are several practical implications arising from our findings for SMEs in cross-cultural environment such as Canada and the UAE. The findings of the study suggest that there is a positive relationship of tacit knowledge sharing with GIC in manufacturing SME, which results in innovation. To promote innovation in manufacturing SMEs it is essential to develop a culture that stimulates individual's tacit knowledge sharing awareness with others. [Crane and Bontis \(2014\)](#) contended that tacit knowledge is an intuitive and subconsciously learned knowledge, which affects our behavior. Hence, SMEs should create processes that support smooth operations and encourage innovation. It is important to have a supportive environment if people need to share tacit knowledge. This is accomplished by building a learning environment that favors openness for sharing what people have learned from one another, including from failure. Managers and academics need to emphasize the tacit knowledge learned from failures not to be silently excluded from explicit knowledge-based focus. Thus, the creation of a learning environment where critical thinking and attempts are made and mistakes are considered a part of learning must be encouraged ([Kucharska and Bedford, 2020](#)). An environment of critical thinking and error acceptance can improve the learning process and innovation ([Kucharska, 2021](#)). This approach needs to be institutionalized within organizations to take advantage of the benefits of error reporting and continuous processing improvement. This study also determines green relational and green human capital as the most important GIC components that facilitate innovation. SMEs need to address the productivity issues associated with these dimensions to enhance innovative capability. Organizational innovation is generated by a firm's tacit knowledge. The development and growth of innovation in the organization depend on intellectual human-knowledge engagement. An environment that can accept failure and supports sharing of knowledge is essential for the survival of the creative spirit and to be ultimately successful.

In order to maximize the potential of GIC, SMEs need to find ways to incorporate AI-based knowledge sharing platforms within their business processes. These systems rise above the realm of classic knowledge management in that they are able to isolate implicit, experience-based knowledge information and transform it into structured, actionable knowledge that filters through the organizational and external channels. For example, SMEs can adopt SAP sustainability management to manage different eco-friendly actions, including reducing the use of resources and cutting their carbon emissions. With these platforms, SMEs can create digital repositories of expertise which can be regularly adjusted with AI's help to deal with new problems in sustainability issues. In addition, by incorporating AI-enhanced collaboration hubs such as Google Workspace AI, the way teams work together can be transformed as they gain predictive insights inside and outside the organization and encourage cross-department shifts and engaging external stakeholders while opening-up sustainability conversations. Successful execution requires a three-pronged phased approach: starting with pilot projects, integrating AI capabilities into the workflow of the organization, and rolling out AI literacy programs to close the skills gap among users. This not only facilitates knowledge transfer but also spreads innovative green practices across the firm.

Sharing tacit information is inherently a social activity. Organizations can boost teamwork and encourage employees to interact with one another which draws new ideas and boosts innovation. Tacit knowledge is accumulated and stored in individuals' mind and can only be transferred when the individual is willing to do so. To encourage "tacit knowledge awareness and sharing," enterprises must understand the

process and develop working environments that facilitate it. Firm level policies to foster innovation should be based on a culture of learning, where colleagues share knowledge obtained from successes and failures. AI significantly promotes tacit knowledge exchange and contributes to enhancing GIC. As the AI databases contain more tacit knowledge, they are more valuable for nurturing GHC and GSC. Leveraging AI can help SMEs to improve their decision-making and innovation capability by producing green products. When SMEs invest in AI for learning and sharing outcomes, they benefit from each other. AI-enabled knowledge management system can help change how tacit knowledge is shared among employees in the firm aspiring to become sustainable. AI applications can record and interpret conversations, documents or designs, exposing that tacit knowledge to others in the organization. By facilitating more efficient and accurate diffusion of knowledge, AI enhances GIC by ingraining green practices in organizational routines, thereby resulting in better green innovation and sustainable product development. For example, a manager uses AI tools to rapidly assimilate complex contextual data from environmentally conscious external consultants and create new, green relational capital and innovation within projects. AI-based collaboration software allows firms to spread tacit knowledge on green actions, improving the shared experience in the organization.

Capturing tacit knowledge, defined as the experience-based insights held by employees, requires a workplace that motivates employees to share their experience. In SMEs, such as the UAE and Canada, the acquisition of tacit knowledge usually starts within trust building and open communication. Methods like job shadowing, mentor programs, and observational learning also aid in knowledge transfer from tacit employees (Nonaka & Von Krogh, 2009). SMEs in the UAE can use interpersonal networks to promote informal knowledge sharing due to the focus on work groups as a result of the collectivist culture of the UAE. Alternatively, Canadian SMEs which are orientated more individualistically in their culture, may employ some of the more structured methods such as interviewing, focus groups or narrative storytelling in order to systematically capture tacit knowledge. Mazuze (2019) argued that technologies like AI enabled tools can help understand verbal and non-verbal cues, transforming tacit understanding into codified wisdom which is available across the firm. The codification of tacit knowledge is the process of capturing subjective insights as operational knowledge in the form of manuals, databases, process maps, or similar. SMEs encounter difficulties in this process due to the lack of resources and the complexity of tacit knowledge that is built on personal experiences and contextual situations. Cultural differences introduce an additional complication to codification; for instance, UAE SMEs might have to deal with high-context communication styles where a great deal of the information is embodied in relationships, whereas Canadian SMEs are in low-context environments that favor explicit and direct written information. To overcome these challenges, SMEs can deploy AI based knowledge management systems to capture and codify patterns from unstructured data, for example, meeting transcripts, or collaborative platforms (Chowdhury et al., 2022). Further, culturally sensitive training built into the process of codification acts as a mechanism to save, or prevent the degradation of, traditional knowledge.

In order for tacit knowledge to be effectively shared within the SMEs, strategies need to be employed that tackle organizational and cultural barriers. Cultural variation among UAE and Canadian SMEs influences the approaches, and the willingness to transfer knowledge. For SMEs in the UAE, there could be a strong preference for in-person interaction, enabled by AI-based team collaboration solutions, that can establish information- and insight-sharing informal networks. Canadian workers are likely to value more structured knowledge resources, such as training workshops or centralized knowledge repositories. Addressing obstacles such as knowledge hoarding and communication lapses can only be achieved by the creation of a culture of knowledge-sharing, incentives for participation, and using technology to overcome geographical and cultural limitations (Raman et al., 2025).

Furthermore, hybrid systems that integrate digital tools and person-to-person contacts can facilitate an effective transfer of tacit knowledge for innovation and organizational development.

Manufacturing SMEs can increase GRC through adopting AI-based collaboration platforms to enable instantaneous interactions and knowledge sharing with external stakeholders, including suppliers, customers, and the experts of environmental protection. This is essential for driving innovation in green products and in sustainability the best practices that are able to capitalize on differing perspectives. AI-enabled analytics can evaluate the environmental performance of supply chain partners and help firms who wish to give preference to partners who care about the environment. This enhances GRC and contributes to the co-creation of environmental products. Setting up AI-powered knowledge hubs encourages collective pooling and sharing of tacit environmental knowledge by internal and external stakeholders, and by doing so it stimulates collaboration and shared sustainability goals. In addition, the AI can be used to arrange virtual eco-innovation workshops where external collaborators are involved in idea generation and problem-solving. AI strengthens green relation capital and catalyzes sustainable innovation projects by curating and processing inputs from such stakeholders. Further, AI's predictability can help businesses uncover nascent green market opportunities both within and outside of existing partnerships for purposes of more proactively co-creation of new products that would enable competitiveness.

RC is crucial for stimulating and enhancing innovation performance. Resources are critical to advance GHC, GSC and GRC as a strategic priority. For a successful improvement in innovation performance, SMEs have to secure sufficient resource allocation to these domains. They also need to invest in promoting socialization, building a supportive learning climate, using AI to enable knowledge sharing, and managing resources strategically on the basis of the goals of organizational innovation. Taken together, these measures will create an atmosphere conducive to innovation and the creation of green human capital. In order for SMEs to maximize their resource allocation to GIC, strategic emphasis is needed to align financial, human and technological resources to sustainability objectives. Investments in AI-enabled training platforms that promote green innovation skill development can also boost GHC by converting employees into sustainability champions. At the same time, investing on predictive analytics tools based on AI, e.g. Microsoft Azure Sustainability Insights, enables SMEs to manage GSC by maximizing supply chain efficiency and reducing environmental impacts. To strengthen GRC, organizations might invest in AI platforms to establish and maintain eco-friendly partnerships, using predictive analytics to pinpoint highly rewarded partners. This allocation of resources should be supported by well-defined quantitative targets, such as a dashboard for monitoring employee engagement with AI initiatives and evaluating the environmental impacts stemming from AI-driven operations.

Manufacturing industries in Canada and the UAE have some specific challenges based on their economic environment and scale of technology utilization. In Canada where sustainable regulation is highly regulated and the labor market is skilled; universities' actions should first concentrate on adapting AI-based technology tools in order to make their operations faster and on logging the compliance with regulations. Advanced predictive analytics and real-time monitoring systems help to improve operations, reduce waste and operate in ways that support green targets. Structured frameworks for capturing and codifying tacit knowledge, particularly through AI-powered knowledge management platforms, can help retain expertise from experienced workers, especially in industries where knowledge attrition is a concern due to an aging workforce. Investment in workforce training programs focused on green technologies and AI applications can address skill gaps and enhance GIC, empowering employees to contribute to eco-innovations and sustainable practices. Furthermore, fostering collaboration among private firms, research institutions, and policymakers can establish networks that facilitate knowledge-sharing and co-innovation, thereby

strengthening the sector's competitiveness.

In the UAE, where rapid industrial growth is paired with a strong emphasis on sustainability, manufacturing firms should concentrate on strategies that integrate AI to overcome inefficiencies and optimize resource utilization. As UAE manufacturing SMEs often operate within a collectivist cultural context, emphasizing collaborative knowledge-sharing frameworks is critical. AI-enabled communication tools and collaborative platforms can foster informal exchanges of tacit knowledge, which are essential for innovation in high-context cultures. Resource allocation strategies should focus on adopting sustainable technologies that align with national initiatives such as the UAE's green growth strategy, ensuring consistency with government sustainability goals. Establishing collaborations with international firms can provide access to the best global practices and advanced technologies, which improves the GRC integration.

Strategic use of AI-based tacit knowledge management and GICs development have wider applications in society, most notably in promoting environmental sustainability and shaping public policy. GIC makes it possible for companies to participate in eco-innovation efforts for solving environmental problems, to mitigate industrial impacts on nature and to stimulate global sustainability initiatives. These approaches are consistent with societal preferences for efficient use of resources, sustainable production methods, and minimization of waste. Moreover, the findings of this study can also provide useful implications for public policies for promoting sustainable practices of SMEs. Such results can inform policy and enable policymakers to create incentives to the AI sector by means of subsidies, or regulations that shall facilitate sustainable innovation. By integrating business strategies and societal needs, these practical measures can contribute substantially to advances in environmental sustainability and economic well-being.

6.3. Limitations and future research

This study is confined to manufacturing SMEs in two distinct countries, Canada and the UAE, which exhibit significant disparities in economic maturity, regulatory environments, and technological adoption. These contextual limitations may influence the generalizability of the findings, as the unique socio-economic and cultural attributes of these regions could shape the dynamics between tacit knowledge sharing, AI integration, and GIC. To broaden the applicability of the results, future research should incorporate SMEs from diverse geographical and economic settings, encompassing both emerging economies and highly industrialized nations. Moreover, conducting longitudinal studies could uncover the long-term effects of AI integration and resource commitment on innovation performance, offering insights into how these relationships evolve under changing technological and economic conditions.

Beyond broadening geographical scope, future research should delve into the sector-specific complexities that shape the integration of AI and GIC. While this study focuses on manufacturing SMEs, sectors such as retail and logistics face distinct challenges and opportunities in adopting AI-driven sustainability practices. For instance, the logistics industry can leverage AI-enhanced supply chain optimization to minimize carbon emissions. Investigating these sectoral nuances would illuminate how industry-specific factors such as regulatory frameworks, market demands, and technological readiness influence the interaction between tacit knowledge sharing, AI, and GIC development. Additionally, exploring tailored strategies, such as sector-specific employee training programs for eco-innovation in high-skill industries or AI-enabled collaborative platforms for fragmented sectors would provide actionable insights to overcome these barriers. By addressing these avenues, future research can enrich the understanding of how AI and GIC drive innovation performance across diverse economic landscapes, enhancing the model's strategic relevance and applicability.

7. Conclusion

Based on knowledge-based view (KBV) paradigm, this study empirically revealed that knowledge tacitness positively influences green intellectual capital, which subsequently supports innovation performance. In knowledge-driven economies such as Canada and the UAE, innovation is paramount for enhancing performance. Thus, as the foundation of innovation, tacit knowledge should be a focal point in both innovation theory and practice. For managers and leaders aiming to cultivate a truly innovative environment within their teams, divisions, organizations, or even at the national level, it is crucial to emphasize and improve tacit knowledge awareness and sharing. Furthermore, the cross-cultural study presented here highlights the significant moderating effect of artificial intelligence on the relationship between tacit knowledge sharing and green intellectual capital within manufacturing SMEs. The study also found that resource commitment enhanced the relationship between green intellectual capital and innovation performance in both Canada and the UAE. Managers and leaders should prioritize tacit knowledge sharing and awareness, leveraging artificial intelligence and resource commitment to maximize innovation outcomes.

CRedit authorship contribution statement

Khalid Mehmood: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Pradeep Kautish:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Resources, Project administration, Investigation, Conceptualization. **Sarah Basahel:** Writing – review & editing, Writing – original draft, Validation, Software, Resources, Methodology, Conceptualization. **Fabio Fiano:** Writing – review & editing, Validation, Resources, Methodology, Investigation, Conceptualization. **Armando Papa:** Writing – review & editing, Validation, Software, Resources, Methodology, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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